Architectural Particular Specification

RAISED FLOOR SYSTEM

SECTION 8.8

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Main Contract for
Proposed Data Centre Development
at 2-10 Tai Yuen Street, Kwai Chung, N.T.
Andrew Lee King Fun & Associates Architects Limited

SECTION 8.8

RAISED FLOOR SYSTEM

PART 1 GENERAL

1.01 SUMMARY OF WORK

A. The scope of work includes design, supply, delivery to site, assembling and installing in position of the raised access floor system at the specified location as indicated on plans.

B. The system provides an access flooring structure comprising of calcium sulphate core covered by corrosion resistant galvanized steel at the bottom with 4 trimmed edges and anti-static high pressure laminate (HPL) on top covering, free-lay onto stringer type steel pedestal understructure fully complied with EN12825 Class 6A Heavy Grade loading performance.

C. The raised floor system shall be supplied and installed in accordance with the manufacturer's recommendations.

D. The Contractor shall detail the raised flooring system including setting-out point and grid-line layout plan that taking into account the system as a whole, including all interface conditions, tolerances and where appropriate, and on-site dimensions.

1.02 REFERENCES

A. The Property Services Agency (PSA) Method of Building Performance Specification 'Platform Floors (Raised Access Floors)

B. Requirements of local statutory authorities.

C. British Standards :

   a. BS 476: Part 4: 1970 (1984)-Non-Combustibility Test For Materials.

   b. BS 476: Part 6: 1989-Method of Test for Fire Propagation for products.

   c. BS 476: Part 7: 1987-Method of Classification of the Surface Spread of Flame of Products.

1.03 QUALITY ASSURANCE

A. The manufacturer of the floor covering must be in possession of a valid quality system certificate showing compliance with ISO9001:2000 quality system.

B. The manufacturer of the floor covering must be in possession of a valid environmental certificate showing compliance with ISO14001 environmental management system.

1.04 PERFORMANCE REQUIREMENTS

A. The raised flooring system shall be installed at locations as shown on the drawings to enable quick and easy accommodation of services lines in any direction. Each panel of the floor shall be interchangeable and allow easy and economical access to the under-floor for maintenance, relocation or new additions of services. The system shall be so flexible so as to enable future extension or additions of single parts.

B. The raised floor system will consist of steel bare access panels, pedestals, stringers and accessories. The system is rated extra heavy duty grade according to Method of Building Standard.

C. Appearance

The completed raised floor system construction shall give a monolithic appearance that all panels shall be flushed without visible steps and square in shape.

D. Technical requirement

1. The raised floor system comprises precision made covered with corrosion resistant galvanized steel at the bottom with 4 trimmed edges and anti-static high pressure laminate (HPL) on top covering, free-lay on adjustable galvanized steel pedestal with stringer understructure to provide a rigid raised access floor above the existing structural floor level. Each panel shall be fully removable and interchangeable with any panel in the system without loss of stability.

2. The raised floor system shall be tested and certified as fully complying with the EN12825 Class 6A Heavy Grade performance specification for raised access floors issued in EN12825:2001 including, but not limited to, the followings:

   a. System Loading

   b. Pedestal Loading

   c. Safety Factor

   d. Hygrothermal Requirement

   e. Fire and Safety Requirement

   f. Electrical Requirement

   g. Acoustic Requirement

3. <u>Panel Construction</u> – Standard size of panel shall be 600 x 600mm, comprises of the fiber reinforced high density calcium sulphate core (not less than 1100kg/m3 - made in Germany by Knauf), covered with corrosion resistant galvanized steel at the bottom and HPL on top covering with 4 trimmed edges of Class 1 self-extinguishing ABS (Acylonitrite Butadiene Styrene) in black color.

   The calcium sulphate core shall be LEED compliance with not less than 40% Pre-Consumer (natural gypsum) and 60% Post-Consumer recycling parts materials, German IBR certified for biological compatible to environment and for low VOC products for indoor application.

   The overall panel thickness shall be 32mm thickness. The floor panels shall be fully interchangeable (with any panels but without loss of stability) and easily cut to adapt to wall and abutments. Panel construction of fully steel encapsulated type and egg-grate bottom type shall not be considered.

   Any equivalent offered shall match or exceed these criteria. The Architect shall have sole discretion in the acceptance and approval of equivalents.

4. <u>Pedestal Material</u> – Pedestal shall be made of solid galvanized steel comprises of 90mm diameter top plate and 100 x 100mm bottom plate. Each plate shall be fully welded to 30mm diameter hollow threaded rods forming an adjustable support of +/-25mm. The pedestal top plate shall be equipped with an anti-vibration gasket with cruciform up stands positively clipped to the steel top plate.

5. <u>Pedestal Fixing</u> – Pedestal shall be fixed to the sub-floor by means of proprietary epoxy adhesive with VOC content less than 30 grams per liter for LEED/BEAM compliance shall not detach from the particular sub-floor when subjected to the PSA MOB sandbag testing for pedestal fixing. The pedestal shall be designed to evenly transmit the static loads over the full area of the sub-floor.

6. Dust-Proof Sealer – Prior to the installation of the raised floor system, the floor void including horizontal floor and vertical wall shall be applied with 1-layer of approved water-based epoxy sealer for prevention of dust accumulation. The sealer shall be LEED/BEAM compliance with VOC content less than 30 grams per liter.

7. Floor Void Insulation – In case for under-floor air-conditioning, a layer of physically (irradiation) crosslinked closed cell polyolefin foam with factory heat sealed reinforced 9um aluminium foil facing with standard size sheet form 1200mm x 600mm x 25mm thickness shall be installed in the void of the raised access floor system including on the sub-floor and at the vertical wall. Insulation material density – 25kg/m3. Thermal conductivity – 0.032W/mk. Fire behaviour BS476 Part 6 & 7 – Class 0.

8. Floor Air Grille – Free area ratio >40% and pressure drop < 20Pa at 700l/s airflow rate & the grille shall be provided with opposed blade.

9. Perforated Panel – In case for under-floor air-conditioning, perforated panel with HPL finish and volume control damper shall be installed at quantity and location as shown in drawings. The minimum design perforation: 304 circular holes. Free airflow: min. 40% (circular holes 24.2mm diameter) and pressure drop < 20Pa at 700l/s airflow rate.

10. Finished Level and Clearance – The finished surface of the raised floor system shall be as indicated in the drawings, and shall be 900mm above the structural floor level. The clearance between the underside of the raised floor system and the structural floor shall be 840mm minimum.

11. Conductivity – The components of the raised floor system shall be conductive. It shall be earth bonded at around one point per 200-250m2 area connected from the pedestal structure to the earth-terminal located outside the meter room at each floor. The continuous electrical conductivity shall comply with the IEE Regulations (18th Edition).

12. Fire Resistance – The raised floor system shall be non-combustible. The raised floor system including the Thermobreak resilient foam edge-seal strip at perimeter shall meet the EN13501-2 fire resistance period of R.E.I. (90 mins), loadbearing capacity, intergrity and insulation.

13. Acoustic Performance – The raised floor system without covering and without barrier in the void shall be designed to achieve room to room sound insulation of normalized flanking sound difference of 52dB and normalized flanking impact sound level of 64dB tested under BS EN ISO 140-12.

14. Cut-Panel Edge Treatment – The burr at the edge of cut-panel shall be removed and then sealed by self-adhesive tin-foil tape for prevention of fire, moisture and vermin attack. A minimum width of 100mm shall be maintained between panel edges and cut-out.

E. Loading Requirement

Loading tests should be conducted on full systems including raised floor panels free lay onto stringer type pedestal understructure adhered to the structural floor by epoxy adhesive at maximum finished floor height FFH900mm:

1. Concentrated load test at the centre, diagonal 70mm from the corner and the weakest edge of panels shall be at least 4.5kN (Working Load) not over 25mm square with maximum 2.5mm deflection.

2. Uniformly distributed load test of the raised access floor system shall be at least 12kN/m2 (Working Load) with maximum 2.5mm deflection.

3. Safety factor shall be 3 times for concentrated load and uniform distributed load without collapse in 5 minutes.

4. Free play pedestal test at pedestal head shall be 5N horizontal load with maximum movement of 1mm per 100mm height.

5. Pedestal horizontal load test at pedestal head at 90Nm for 5 minutes as per applied load range 9kg to 50kg with maximum deformation 1mm per 100mm height.

6. Pedestal vertical load test over pedestal centre and one quadrant head shall be 18kN and 13.5kN respectively for 5 minutes without collapse.

The dead load of the system shall not exceed 56kg/m2 for and weight of each panel shall not exceed 19.5kg.

F. Hygrothermal Requirement

The Floor System, including the pedestals, panels, floor coverings and fixings shall be capable of complying with the performance requirements stated in PSA MOB PF2/SPU medium grade standard without any delamination or other forms of deterioration when subject to the following hydrothermal conditions :

1. Temperature 5°C, relative humidity 90 % on both side of the floor panels.

2. Temperature 50C and 95% RH in the floor void side of the panels. Temperature 250C and 25% RH on the room side of the panels. The Contractor shall state extent of movement under the above-mentioned conditions, and state what effects such movement will have on the system.

3. The material content of the floor panels shall not be capable of absorbing moisture.

1.05 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

A. The Contractor and Raised Floor System Manufacturer shall warrant that all products, materials and items are suitable, and can be maintained in the various climatic and psychometric conditions encountered without degradation, material and support failures.

1.06 SUBMITTALS

A. Submit the following under provisions of Section 1.5.

B. Manufacturer's catalogues, test certificate, and installation method, material samples and shop drawings for all relevant materials and equipment shall be submitted to the Architect for review or approval. The extent of information or samples required shall be adequate and sufficient to demonstrate that the proposed system and materials are in compliance with the contract requirements. However, the Architect can request the contractor to submit any additional or supplementary information to substantiate the performance of the proposed system or material.

C. Shop Drawings:

1. Indicate profiles, sizes, connection attachments, reinforcing, anchorage size and type of fasteners and accessories.

2. Include erection drawings with plans and details where applicable. Layout and plan drawings shall be drawn at 1:50 scale. Detail drawings at 1:1 scale.

3. Indicate location, and size of all cut tiles.

D. Samples of each type of access floor including floor grille. Samples of cut tiles to be submitted for review.

E. Operation and Maintenance Manual

The Contractor shall submit a copy of the manufacturer's operation and maintenance manual for approval by the architect, upon the completion of the raised floor works. This manual shall include the followings:

1. Installation details, technical brochures, material list and all accessories which have been installed/added following completion of the raised floor works.
2. The correct method of lifting and replacing panels and any stringer, including limitations on the sequence and number of panels, stringers, etc. which can be removed at any time during servicing without endangering the safety of the floor system.
3. The permissible loading that can be applied to the floor, with guidance on the use of spreader plates, etc. during the installation of equipment and subsequent maintenance.
4. Methods of dismantling and re-fixing of raised floor panels by others for installation of walls, partitions, cabling, ducts, etc. This is to ensure no damage of raised floor system and no rocking of panels causing squeaking sound.
5. Cleaning method for the panels and any integral finishes.

F. As-fitted Drawings

On completion of the installation and following receipt of the architect's approval of the as-fitted drawings and maintenance manual submissions, the Contractor shall provide to the architect, two prints and one reproducible copy of each as-fitted drawings as well as two copies of bound manuals incorporating the catalogue, technical information and maintenance manual.

1.07 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.08 DELIVERY, STORAGE AND HANDLING

A. Materials shall be delivered in Raised Floor System Manufacturers' containers, dry, undamaged, unopened packages, all clearly labelled with the manufacturer's name, product identification, and lot numbers intact.

B. Materials shall be stored in strict accordance with the Manufacture's printed instructions, copies of which will be furnished to the Architect.

C. Protect access floor panels from staining, discoloration and damage.

D. Inspect all products for defects and damage. Return and replace all damaged items.

E. The Contractor shall be responsible for protecting access floor panels and finish accessories against (physical and climatic) during storage.

F. Panels, pedestals, stringers, gaskets and ancillary items shall be packed separately.

G. The delivered product shall be stored, installed, used and maintained in dry, closed, ventilated place, not subject to quick variations of temperature and humidity, with temperature between 5°C and 35°C and relative humidity between 40% and 70%.

1.09 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.10 TESTING

Should the architect so required, the contractor shall arrange sand bag test for pedestals selected at random at the rate of 0.5% of the number of pedestals for each floor. Testing shall be carried out as described below and under approval and supervision of the architect.

The test shall be commenced until at least 48 hours have elapsed from the time that any adhesives, used to install the pedestal were first applied.

A soft body comprising a canvas bag of diameter 150 x 250mm containing dry sand and weighing 3 kg shall be arranged to swing through and arc of 90 degrees at radius of 1 meter to strike the head of the selected pedestal. When subjected to this dynamic force, the pedestal structure shall not fail and the pedestal shall remain firmly fixed to the sub-floor.

1.11 WARRANTY

A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

B. The works contractor shall furnish a material warranty to the Employer for a minimum maintenance-free period of five (5) years for use of the entire raised floor.

C.

1.12 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 APPROVED MANUFACTURERS / SUPPLIERS

A. Subject to compliance with the Project Contract Documents, provide raised access floor system and all necessary accessories from one or more of the following approved manufacturers / suppliers, or equivalent approved by the Architect.

1. Shun Wah Contracting Co Ltd.
   Room 803, 8/F, Block A, Sea View Estate, No. 2 Watson Road,
   North Point, Hong Kong
   Kitty Cheng
   Tel: (852) 2898 2378 Fax: (852) 2898 1733
   E-mail: kitty@swc.hk

2.02 MATERIALS

A. Access Panels

High density fibre-reinforced calcium sulphate core, protected by fire retardant edge trim bonded to all four edges, with galvanized steel bottom and finished surface is HPL / PVC / Bare. Nominal size 600mm X 600mm, thickness no less than 31.5mm.

B. Pedestals

1. Each pedestal consists of:-
   a. flat steel base 100mm x 100mm, thickness no less than 2.5mm,
   b. steel tube by approximately 31.8mm outside diameter,
   c. head consisted of a circular steel plate approximately 90mm outside diameter.
   d. head tube mated with the steel collar to provide height adjustment and is locked with a steel locking nut.
   e. a 2.5mm thick plastic pad with moulded downwards fitted over the pedestal head.
2. The pedestal system shall be corrosion resistance by zinc plating, no less than 5 micron or hot dip galvanized steel.
3. Loading:-
   a. Pedestal shall pass PSA MOB PF2 PS SPU (UK) – Medium Grade performance specification independent test.

C. Stringers

The stringer shall be made from a steel box channel approximately 540mm (L) x 25mm (W) by 25mm (H), with thickness no less than 1.5m. Both the free end are tapered with a circumferential slot of 8.3mm (L) x 5.7mm (W) near each free end.

D. Accessories

Adhesive epoxy, steel stringer, floor air grille, perforated panel, fascias, entry ramp, multi steps and glass display panels.

1. Finish on the panels and the system can be in:-

   a. Bare finish, carpets should be installed on top of the system; or

   b. High Pressure Laminate (HPL) finish, HPL panel shall be corrosion resistance treatment by powder-coating. 1.2mm thick HPL to be factory bonded on the surface with conductive edge around four sides.

      High Pressure Lamination (HPL) panel surface Performance:

      Antistatic – 1.0 x 10^8 ~ 1.0 x 10 ^11ohms; or

   c. Vinyl finish, 2.0mm / 3.0mm / 4.5mm vinyl flooring, to be factory bonded on the surface with conductive edge trims around four sides

E. Perforated Panel

1. Steel Panel and understructure in steel grid welded, nominal size 600mm X 600mm thickness no less than 34 mm. The panels shall be corrosion resistance treatment by powder coating. 1.2mm thick HPL finish to be factory bonded on the surface with conductive edge trims around 4 sides.

2. Concentration load property of the access panel, by 1" plate, shall be greater than 1000 lb (4.0kN) at less than 2.5mm depression.

3. Concentration ultimate Load of the access panel: Shall be no less than 3,000lbf (13.0kN).

4. Holes

   22mm diameter x 400 numbers

5. Open ratio: approx: 42%

6. Finish Materials on Raised Floor System

   a. High Pressure Laminate (HPL)

      1) Panels Size: 600mm x 600mm

      2) Thickness: HPL: 1.2mm

      3) Thickness of Access panel: 36mm

      4) Static Resistance at 1.0 x 10^8 ~ 1.0 x 10 ^11ohms

   b. All HPL panels shall be pre-bonded in raised floor manufacturer’s factory.

   c. Borders of the HPL panels shall be protected by conductive edge trims.

2.03 ANCILLARY MATERIALS

A. Gasket for perimeter cut panels and around pillars

Self-adhesive expansion gasket shall be applied to the cut panels along the walls/pillars to prevent air leakage and movements of the floor.

B. Ramps and Stairs

Internal or external ramp with black no slip rubber finish, complete with substructure, perimeter upstanding and aluminium profiles.

C. Openings

Openings in floor panels shall be protected with self-adhesive tin-foil tape at cut-edge and allowed for service outlet floor box and cable outlet grommet with removable cover cap.

D. Panel Lifting Device

Supply at least 3 nos. of panel lifting devices for easy dismantling and re-fixing of the floor panels during maintenance. Each lifting device shall have a handle for easy griping.

All panels shall be capable of being removed and replaced without the use of undue force by the panel lifting device. The lifting device, when used in accordance with the manufacturer's instruction, shall not cause damage to the covering material bonded to the floor panel, nor causing any failure of the bond.

PART 3 EXECUTION

3.01 EXAMINATION / INSPECTION

(NOT USED)

3.02 PREPARATION

A. All cracks, holes and imperfections of the sub-floor shall be patched.

B. Dust proof sealer

1. The concrete floor slab shall be generally level and not exceeding the allowed building tolerance.
2. The surface of this concrete sub-floor shall paint an approved sealer before the installation of the raised floor system.
3. The sub-floor to be clean, dry and free from dust, dirt and other contaminants.
4. A color tined sealer shall be applied to all concrete surface within the floor void to prevent dusting throughout the life of the installation

3.03 INSTALLATION

A. The system shall be installed in accordance with manufacturer's guidelines and installation manual.

B. The sub-floor for installation must be smooth, clean, dry and structurally sound. If direct-to-earth installation is required, an effective waterproof membrane must be incorporated. A moisture test must be carried out to ensure that the subfloor has dried out to a level consistent with the application of sealer.

C. Installation shall not be conducted if in-door temperature of the job site falls lower than -5 degree centigrade (23 Fº)

D. Sealer / Dustproof

a. The concrete floor slab shall be generally level and not exceeding the allowed building tolerance.
b. The surface of this concrete sub-floor shall paint an approved sealer before the installation of raised floor system.
c. The sub-floor to be sealed shall be clean, dry and free from dust, dirt and other contaminants.
d. A colour tinted sealer shall be applied to all concrete surface within the floor void to prevent dusting throughout the life of the installation.

E. Tolerance

1. The completed raised floor system shall be rigid, flat and free from vibration, rocking and squeaking sound. The Contractor shall provide equipment and specialist for laser level checking for raised flooring handover. The laser level checking shall be carried out under approval and supervision of the Architect. The overall floor shall be level to within:
   1) +/- 1.5mm over any 5 m2
   2) +/- 6.0mm over the entire floor
2. The Contractor shall employ only specialist installers that are thoroughly experienced with this type of installation.

3.04 CLEANING

According with the instruction supplied by the access floor manufacturer.

On completion of the raised floor works, the contractor shall remove the protection and final clean the raised floor system for handover to the employer.

3.05 PROTECTION

According with the instruction supplied by the access floor manufacturer.

Immediately following the completion of the installation at each phase, the raised floor system shall be protected by 2mm thickness hollow plastic board to prevent dust and dirt contamination as well as for protection against inspection and minor patch-up works prior to handover to the employer.

During and after completion of the work under this section, it shall be the responsibility of the Contractor, to ensure that the work is properly and adequately protected from damage.

END OF SECTION